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**Analysis of ultrasonic vocalization in mice for the
diagnosis of autism**

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List of abbreviations

ASD - autism spectrum disorder
mASD – mouse autism spectrum disorder
USV - Ultrasonic vocalizations
MTHFR - methylenetetrahydrofolate reductase
PANNs - Pretrained audio neural networks
WT – Wild gene (no ASD)
HT – infected gene (with ASD)
TP – True positive
TN – True negative
FP – False positive
FN – False negative
STFT - short-time Fourier transformation

1 Abstract

Autism Spectrum Disorder (ASD) is a multifaceted developmental condition shaped by both genetic and environmental factors. Early detection and intervention are critical for enhancing behavioral outcomes in affected individuals, yet current diagnostic methods often rely on subjective observations of behavioral abnormalities, which can be time-consuming and lack objectivity. This underscores the need for an objective detection tool. Mice, known for their social behavior, serve as valuable models in autism research. These rodents produce ultrasonic vocalizations (USV) during social interactions, providing a unique opportunity to explore the biological mechanisms underlying atypical social development in ASD. In this study, we propose the development of a machine-learning-based model that analyzes USV signals to differentiate between healthy mice and those displaying ASD-like behaviors. Our approach integrates an established segmentation and classification model with novel features specifically tailored for ASD detection. This research has demonstrated that machine learning models, particularly those leveraging the analysis of USV. The findings suggest that the USV classification by majority voting method, based on the study of S. Weinberg and Ben-Gurion study ,although challenged by data bias and limited testing samples, achieved a moderate accuracy of 62%. Meanwhile, the Pretrained Audio Neural Networks (PANNs), especially when focused on the genetic background of the mice, showed promising results, with accuracy rates as high as 82% when analyzing the gene of the mother. However, the results were inconsistent, particularly when analyzing the gene of the infant, where significant discrepancies were noted. The project highlights the potential of using advanced audio processing techniques combined with machine learning for early ASD detection, while also pointing out the need for further refinement and testing to address observed result discrepancies and improve model robustness.

2 Project Goals

Detecting ASD in mice is a challenging task that requires precise and reliable methods. In this study, we address the limitations identified in Weinberg’s study, specifically the lack of subject-independent identification, by incorporating two novel methodologies:

1. Majority Voting using XGBoost (a subject-level approach).
2. Pretrained Audio Neural Networks (PANNs, a segment-level approach).

These approaches reduce the risk of data leakage and enhance the reliability of ASD classification by considering predictions from multiple segments of the same mouse and utilizing advanced audio processing techniques.

3 Introduction

3.1 Understanding and diagnosing ASD

ASD is a neurodevelopmental condition marked by a wide range of symptoms, primarily in communication and social interaction, alongside repetitive behaviors [1][2]. Affecting about 1 in 68 children, ASD's early identification is crucial for timely intervention, which can significantly enhance life quality for affected individuals [1]. Diagnosing ASD in children often relies on observing atypical behaviors, but the variability in symptoms means that diagnosis usually occurs around 4-5 years of age, despite reliable methods being available for severe cases before age two [2][3]. Early detection is critical, as interventions started before age four can lead to substantial improvements in various developmental areas [1]. Recent studies have explored early signs of ASD in infants, particularly in those at high risk, such as siblings of diagnosed children. These infants may show delays in speech-like vocalizations and other early developmental markers, suggesting that signs of ASD can manifest before verbal speech emerges [4][5]. Identifying these early indicators is essential, and controlled genetic animal models are valuable tools for understanding the genetic factors that contribute to autism.

3.2 MTHFR and autism spectrum disorder

The potential of early detection is enhanced by insights into genetic influences, including the role of the methylenetetrahydrofolate reductase (MTHFR) gene, a key player in the C1 metabolic pathway. Research using mouse models has demonstrated that a deficiency in MTHFR is associated with developmental delays and symptoms indicative of autism, shedding light on the genetic underpinnings of the disorder [6].

3.3 Exploring autism spectrum disorder through mouse USV

Research involving rodents, particularly mice, has explored specialized sounds within the ultrasonic range known as USV. These vocalizations are emitted in various social contexts [7]. Utilizing the mouse model of autism provides valuable insights into the biological mechanisms underlying abnormal development observed in ASD and informs potential treatments [6]. In mouse pups, the production of USV is influenced by factors such as maternal isolation, cold exposure, rotation, and genotype. Consequently, the standard test for vocalizations in mice involves assessing the ultrasonic distress calls emitted by pups separated from their mothers or removed from the nest [8] [9]. These calls serve to investigate the number of calls and the types of syllables emitted by separated pups in mouse models of autism spectrum disorders [9]. The behavioral analysis of USV in mouse models of ASD provides a foundation for understanding the nuances of mouse communication. Mice employ various communication methods. They emit audible squeaks for long-distance warnings, falling within the human-audible range. Additionally, for short-distance communication, mice use USV with frequencies spanning from 25 to 120 kHz. These USV exhibit considerable variability and complexity. They consist of distinct syllable types, with temporal sequencing involving the repetition of phrases [8]. Researchers often categorize mouse USV into ten different syllable types based on differences in

duration, primary frequencies, and spectrograms (e.g., flat, short, upward, downward, etc.) [9][10].

3.4 Advancements in mice USV syllables classification

In recent years, attempts have been made to classify syllables into fewer and more general categories as well as to classify using software, in order to improve the ability to detect alterations in model pups [11]. We will review some of the notable studies and methodologies. The summary table indicates that DeepSqueak has achieved the highest level of accuracy in classifying mouse USV syllables, with an impressive 95% accuracy rate.

Table 1 - USV classification tools comparison

	MUPET [12]	DeepSqueak [13]	VocalMat [14]	BGU Project [15]	Joseph the MoUSE [16]
Year	2017	2019	2021	2021	2023
Main method	K-mean clustering	Faster RCNN	CNN	CNN	FRCNN GAC
USV database size	51,000~	56,000~	45,000~	~6000	Unknown
Syllable categories number	60-200 Depend on user-defined	100 Blink to the smallest possible number of categories	11	others+10	6
Average accuracy result	85%	95%	86%	81%	79%
Limitation	Clustering is applied on the spectral magnitude of the segmented syllables so that amplitude changes can impair classification to the different categories	It may be classified into a small number than wanted of categories	Shows lower accuracy for detecting USV with multiple components	Small training dataset	Fewer categories and less accuracy than other tools and studies

3.5 Ben-Gurion University work

Building on the segmentation and classification of USV calls, Shaked Weinberg's study [17] implemented a machine-learning model specifically designed to analyze the USV of mice to determine whether a mouse is healthy or shows ASD-like symptoms. This approach facilitates error detection during the learning process and is particularly effective in structured data scenarios. The model operates as an end-to-end system, processing the entire USV signals from segmentation to final diagnosis without the need for manual intervention. It relied on features extracted from the USV signals, which were selected based on a

review of existing literature and previous studies, and included various vocalization characteristics such as frequency, duration, and syllable types.

3.6 Utilizing PANNs for mASD detection

In addition to Shaked's study [17] on classifying mouse USV syllables, researchers have explored using PANNs [18] for detecting mouse models of ASD. PANNs, trained on the extensive Audio Set dataset, applied to various audio-related tasks. A notable example is the Wavegram-Logmel-CNN architecture, which combines both log-mel spectrogram and waveform as input features. This method has achieved state-of-the-art performance in audio tagging tasks, outperforming previous systems [19].

4 Methods

4.1 Dataset

The dataset comprises recordings of 127 mice, categorized by offspring and maternal genotypes across five years (2015, 2018, 2022, 2023, and 2024). The most substantial group consists of 55 WT offspring from WT mothers, followed by 44 WT offspring from HT mothers, and 28 HT offspring from HT mothers. In total, 29,861 recordings were collected, with the highest number in 2015 (10,391 recordings) and the lowest in 2023 (3,020 recordings). The dataset reveals significant variation in the number of recordings per year and genotype combination. This dataset provides a foundation for evaluating the PANNs model's performance in classifying mice audio recordings. To ensure a robust and unbiased evaluation, we split the dataset into training, validation, and test sets with careful separation of mice.

Table 2 - Mice's and audio recording data

Offspring Genotype	Maternal Genotype	2015	2018	2022	2023	2024	Total
HT	HT	10	4	7	5	2	28
WT	HT	18	7	4	9	6	44
WT	WT	10	8	13	8	16	55
Total Mice		38	19	24	22	24	127
Number of Recordings		10391	6971	3839	3020	5640	29861

4.2 USV classification by majority voting

4.2.1 Background

The dataset used in the study was collected from Golan’s laboratory and consists of 7923 audio recordings of 70 young mice, which include both healthy mice and those with MTHFR gene deficiency. The USV were recorded with a frequency range of 20 kHz to 200 kHz, and the sampling frequency was set to 250 kHz to capture the highest frequency of USV, which is around 120 kHz. The model achieved an average classification rate of 79% for identifying ASD-like signs in mice, indicating its effectiveness in distinguishing between healthy and affected individuals based on their vocalization patterns. The final results of the study indicated a relatively high classification rate for identifying ASD-like signs in mice, achieving an average accuracy of 79%. The features used in the model were derived from the USV signals, focusing on characteristics such as frequency, rhythm, intensity, and repetition. The conclusions drawn from the results highlighted the significance of the mother's genotype as a crucial feature in the classification process. However, the study also noted issues related to the small number of mice in the dataset, which could lead to instability in the model's performance when identifying ASD signs on a per-mouse basis rather than per recording. The research emphasizes the importance of genetic factors in understanding ASD, suggesting that further exploration of gene-related features could enhance the model's predictive capabilities. We acquired the original data and code used in the previous project and executed it end-to-end in our study to establish an identical starting point.

4.2.2 Our starting point

In previous research on the classification of ASD in mice, the methodology involved shuffling mouse samples. This approach caused data leakage, as segments from the same mouse could appear in both the training and test sets. This leakage resulted in biased results because the model could indirectly "see" the test data during training, leading to over-optimistic performance estimates. To solve this problem, we implemented subject separation, ensuring that data from each mouse is either entirely in the training set or the test set, but not both. While this approach eliminates data leakage and provides a more realistic evaluation of the model's performance, it initially led to a decrease in performance due to the more stringent testing conditions. To address this, we introduced a majority voting algorithm that aggregates predictions from multiple segments of the same mouse. This method improves the robustness and accuracy of the classification by considering the consensus among multiple predictions rather than relying on a potentially biased single-segment prediction. However, this solution comes with a trade-off. By aggregating multiple samples into a majority decision for each mouse, the effective number of individual predictions is reduced, which could potentially impact the study’s ability to detect subtle patterns in the data. Despite this, the benefits of eliminating data leakage and achieving more reliable and unbiased results are considered to outweigh the potential drawbacks of having fewer individual predictions.

4.2.3 Pre-processing

We applied pre-processing methodology to mice recordings collected from 2015 to 2022, comprising approximately 20,000 recordings. Building on the previous work done by BGU [15] we utilize a pre-processing pipeline to extract syllable segmentations with precise time and frequency start and end points. The pre-processing pipeline involved several key steps. First, the raw audio data was segmented into individual syllables. For each syllable, we determined the start and end frequencies, as well as the duration of the syllable. Additionally, each syllable was classified into 1 out of 10 possible syllable classes, we are using the same features that was used in the previous work [17].

4.2.4 Dataset Split with Mice Separation

To ensure a robust and unbiased evaluation, we split the dataset into training, validation, and test sets with careful separation of mice:

- Training set: A portion of the data was grouped by mice index and then splited according to mASD and WT, allocating part of the USV data from each mouse to the training set, which was used to train the XGBoost model.
- Validation set: Another portion of the data was allocated to the validation set, which was used for hyperparameter tuning and model validation.
- Test set: The remaining mice were allocated to the test set, which was used to evaluate the final performance of the model. This split ensures that the test set provides an unbiased estimate of the model's performance on unseen data.

4.2.5 Classification Using XGBoost

Each segment of the audio recordings is processed independently to assess the likelihood of ASD-like behaviors in mice. The process includes:

- Model Training: Train the model on the extracted features, ensuring the model learns to distinguish between ASD and non-ASD behaviors based on the segment-level features.
- Cross-Validation: Use cross-validation to tune the hyperparameters of the model.

4.2.6 Majority Voting Algorithm

To enhance the reliability and validity of the ASD classification, we implement a majority voting algorithm:

- Independent Segment Assessment: Each segment was independently assessing the likelihood of ASD-like behaviors using the trained model.
- Consensus Classification: The final classification for each mouse is determined by the consensus among the predictions from multiple segments. Specifically, the majority vote (i.e., the class with the most votes) is taken as the final decision.

4.2.7 Evaluation and Optimization - Performance evaluation

Finally, to evaluate the performance of our classification model, we implemented a cross-validation strategy using k-fold cross-validation, where each fold contained different mouse subjects. This approach ensured that data from the same mouse did not appear in both the training and validation sets within a fold, thereby preventing data leakage and providing a more realistic assessment of the model's generalization ability. Furthermore, we used the cross-validation process to tune the model's hyperparameters. Specifically, we optimized the number of estimators for our classifier by selecting the parameter value that yielded the highest average accuracy across the k folds. This systematic approach ensured that we identified the most effective model configuration before evaluating its performance on the final test set. We evaluated our model using accuracy. Accuracy provides a straightforward measure of the proportion of correct predictions.

4.3 Voting Results

The dataset used for this experiment was distributed with a 30% majority of mASD mice and 70% wild-type (W) mice. A total of 69 mice were included in the study, with 20 mASD mice and 49 W mice. The data was split into training and validation sets, with 70% (48 mice) used for training and 30% (21 mice) for testing. A 4-fold cross-validation was employed to assess the robustness of the model, with the results analyzed using a confusion matrix.

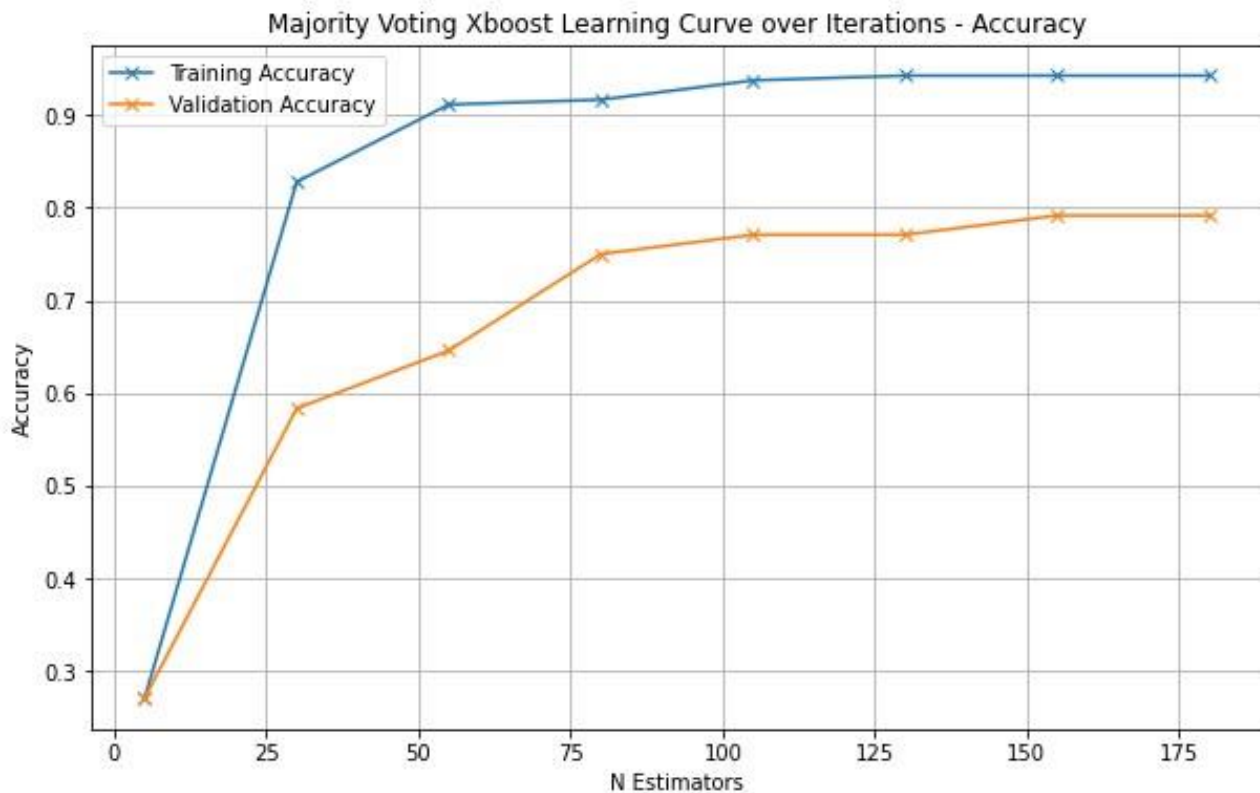
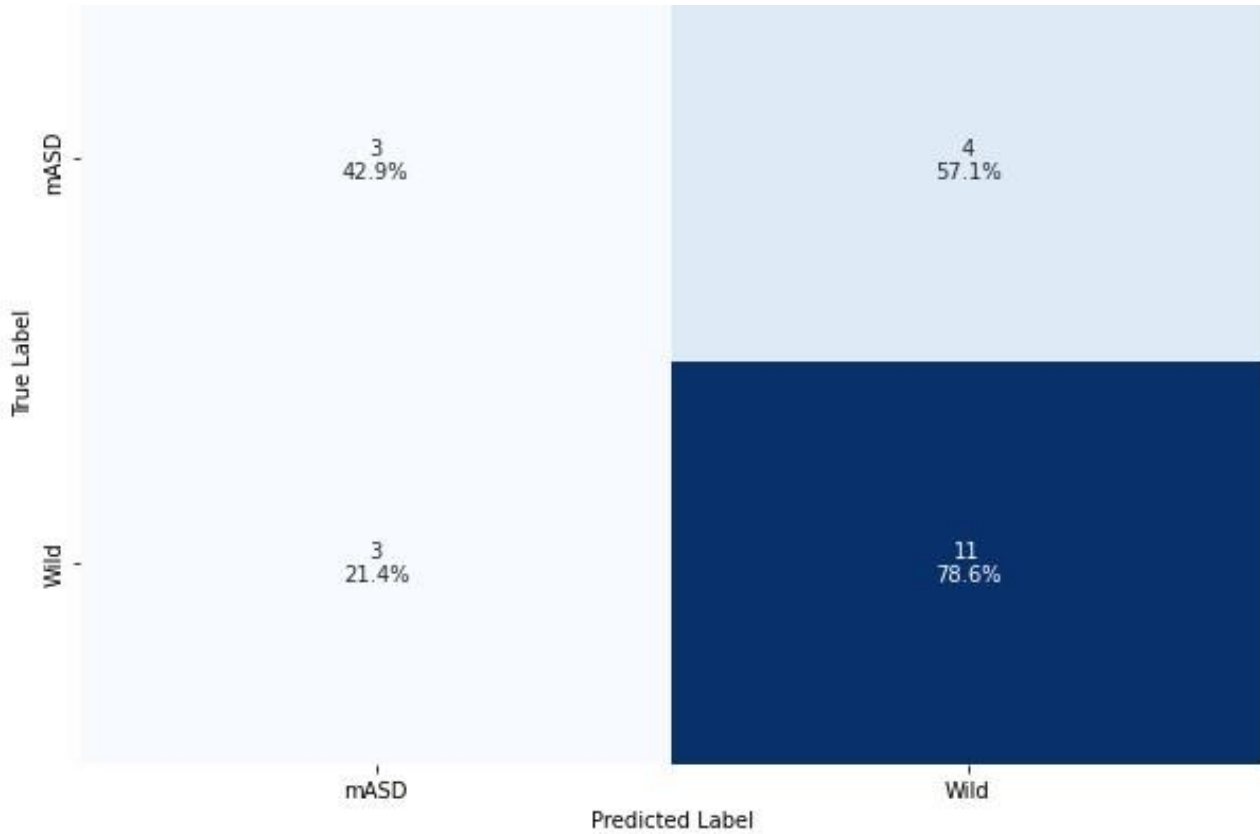


Figure 1 - Majority voting learning curve

The learning curve analysis of the XGBoost model applied to classifying mice with ASD reveals that the model exhibits signs of overfitting. Training accuracy rapidly increases and stays at

a high level, while validation accuracy improves more gradually and stabilizes around 75 estimators. This indicates that further increasing the number of estimators beyond 75 does not yield significant improvements in validation accuracy. The optimal number of estimators is likely around 75, balancing model complexity and generalization.

Table 3 - Majority voting test results



4.3.1 Overall model performance for majority voting

The model is more accurate in identifying Wild mice than detecting mASD hence fails to perform the binary classification , with a TN (78.6%) TP (42.9%). FN (57.1%) FP (21.4%) in general the classification results are not promising also highlighting challenges in detecting mASD.

4.3.2 Interpretation discussion and recommendations

The accuracy achieved with the majority voting method was 0.66. Several limitations affected the performance of this method:

- Limited Number of Mice in the Test Set: With only 21 mice in the test set, the sample size was too small to provide a robust evaluation of the model's performance. This limited the ability to improve accuracy significantly.
- Bias Towards ASD Mice: The dataset was biased towards mASD mice, making it difficult to generalize the findings to a broader population. This imbalance likely contributed to the model's decreased accuracy on the test set.

- Inability to Improve Results with Augmentation: Despite efforts to enhance the dataset through augmentation techniques, the model's accuracy did not improve, suggesting that the existing data and model parameters might not be sufficient to overcome the limitations posed by the dataset's characteristics. Overall, while the majority voting method demonstrated some potential, the results highlight the need for a more balanced dataset and possibly more advanced techniques to improve the model's generalizability and accuracy.

4.4 PANNs

In this section we describe a different approach based on [18]. We will apply PANNs to the processed signals to improve estimations. This strategy aims to enhance the accuracy of our ASD detection by fine tuning the pretrained model.

The PANNs approach leverages the CNN-14 architecture within the Wavegram-CNN systems to address limitations of previous one-dimensional CNN models, which struggled to capture crucial frequency information in audio signals. Traditional one-dimensional CNNs primarily processed audio in the time domain but failed to effectively capture frequency patterns, essential for accurate audio pattern recognition. To overcome this, the authors of the PANNs paper introduced the Wavegram-CNN and Wavegram-Logmel-CNN systems. These systems aim to learn a time-frequency representation directly from raw audio waveforms, creating a feature called a Wavegram that captures both time and frequency data. By combining the Wavegram with the traditional log mel spectrogram, the Wavegram-Logmel-CNN system leverages the strengths of both learned and hand-crafted features to enhance audio tagging performance [19].

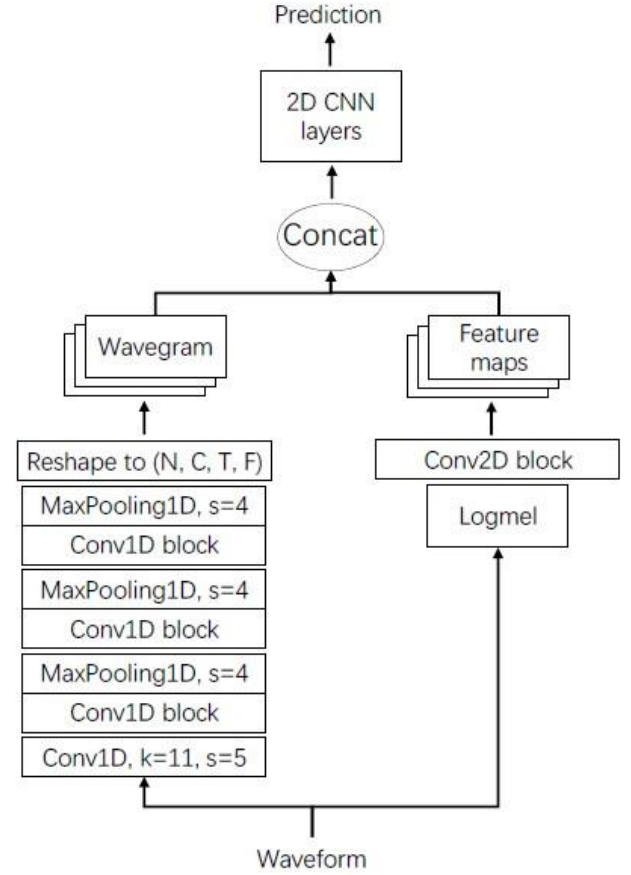


Figure 2 - Wavegram-Logmel-CNN architecture

In a specific application of PANNs, a study was conducted to analyze USV from mice to detect mASD. The study utilized a dataset of USV from 168 mice, with recordings taken at postnatal days 8 and 12. These recordings, captured at a sampling rate of 300 kHz, were analyzed using STFT to explore the time-frequency domain. The large-scale PANN model, specifically designed for audio pattern recognition, was employed to process these vocalizations, with a focus on the vocalizations themselves rather than the genetic background of the mice. The study achieved a UAR of 79.2% in detecting mASD in a subject-independent scenario, highlighting the potential of PANNs in this context. However, the study also noted limitations, including the need for more explainable models and further exploration of traditional acoustic features [19].

4.4.1 Data preprocessing

To prepare the collected audio data for training a neural network, a series of preprocessing steps were undertaken:

- **Data Organization:** raw audio files were reorganized into a hierarchical folder structure. Each folder corresponded to a specific class:
 1. **Two-Classes:** The data was categorized into two classes based on the genotype: Wild Type (WT) vs. Heterozygous (HT).
 2. **Three-Classes:** The data was further refined using the mother's genotype, resulting in three classes: WT_WT (Wild Type offspring from Wild Type mothers), WT_HT (Wild Type offspring from Heterozygous mothers), and HT_HT (Heterozygous offspring from Heterozygous mothers). This deeper categorization aimed to capture more precise genetic influences on the audio signals.
- **File naming and metadata management:** consistent file naming conventions and metadata management were implemented to ensure traceability and ease of access. This step involved renaming audio files to reflect their respective classes and maintaining a metadata log to track additional information about each recording.
- **Data augmentation:** To enhance the robustness of the neural network, the 'mixup' data augmentation method was applied to the audio files. This technique involves creating new training examples by combining pairs of samples with a certain mixup ratio. By blending the audio signals and their corresponding labels, mixup helps improve the model's generalization ability and reduces overfitting.

4.4.2 Data usage and limitations

Due to memory constraints, we utilized 50% of the available data for the PANNs methodology. Specifically, this involved selecting 55 recording files from each mouse. This limitation was necessary to ensure that the data could be processed and analyzed effectively within the available computational resources.

4.4.3 Frequency configurations

The frequency configurations are critical for transforming audio signals into a format suitable for PANNs:

- **Sampling Rate:** The sampling rate was set to 250,000 Hz.
- **Clip Samples:** The number of samples in each clip was defined as $\text{clip_samples} = \text{sample_rate} * 3$ (i.e., 750,000 samples per 3-second clip).
- **Spectrogram Conversion:** Audio signals were converted into spectrograms, representing the frequency spectrum over time. Key parameters included:
 - **Mel Bins:** 64 mel bins were used to represent the frequency spectrum.
 - **Frequency Range:** The minimum and maximum frequencies were set to 20,000 Hz and 120,000 Hz, respectively ($f_{\min} = 20000$, $f_{\max} = 120000$).
 - **Window Size:** A window size of 1024 samples was used for the Fourier transform ($\text{window_size} = 1024$).

- Hop Size: The hop size (the step between successive windows) was set to 320 samples (hop_size = 320).
- Window Function: A Hann window was applied to each segment (window = 'hann').
- Padding Mode: 'Reflect' padding mode was used to handle the edges of the audio signal (pad_mode = 'reflect').
- Centering: The window was centered on the signal (center = True).

These are parameters yield the best results.

With the processed audio signals, PANNs were applied to improve ASD detection:

- Pretrained Model Selection: A pretrained PANN model trained on a diverse set of audio data was selected, ensuring it could generalize well to our specific task.
- Fine-Tuning: The pretrained model was fine-tuned on our dataset to adapt it to the specific characteristics of mice vocalizations. This involved:
 - Transfer Learning: The PANN was initialized with the weights of the pretrained model.
 - Training: The PANN was trained on our processed audio signals, using a labeled dataset of mice vocalizations with known ASD diagnoses.

Adapting the classification output of the PANN to our dataset involved the output layer modification: The output layer of the PANN was adjusted to match the number of classes in our dataset. In our case it could be 2 or 3 classes. The performance of the PANNs model was evaluated by accuracy, this metric measures the proportion of correctly classified instances out of the total instances. It provides an overall assessment of the model's ability to correctly identify the classes in the dataset.

4.5 PANNs results

Due to memory limitations, we restricted the number of audio files per mouse to 50. The dataset was divided into train, validation, and test sets with no overlap between individual mice. This division ensures that the evaluation is robust and that the model's performance is assessed on entirely unseen data. The test set included 12 mice, providing a challenging scenario for the model's generalization capabilities.

4.5.1 Mother gene results

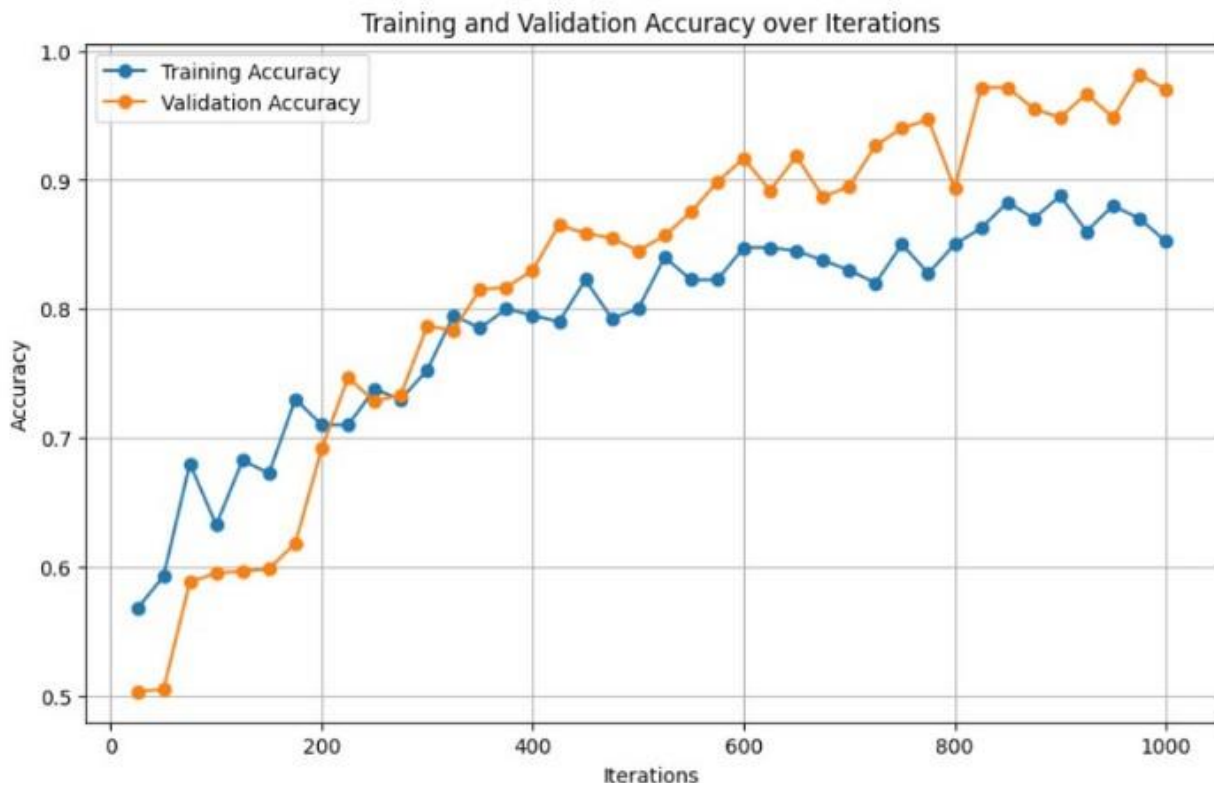
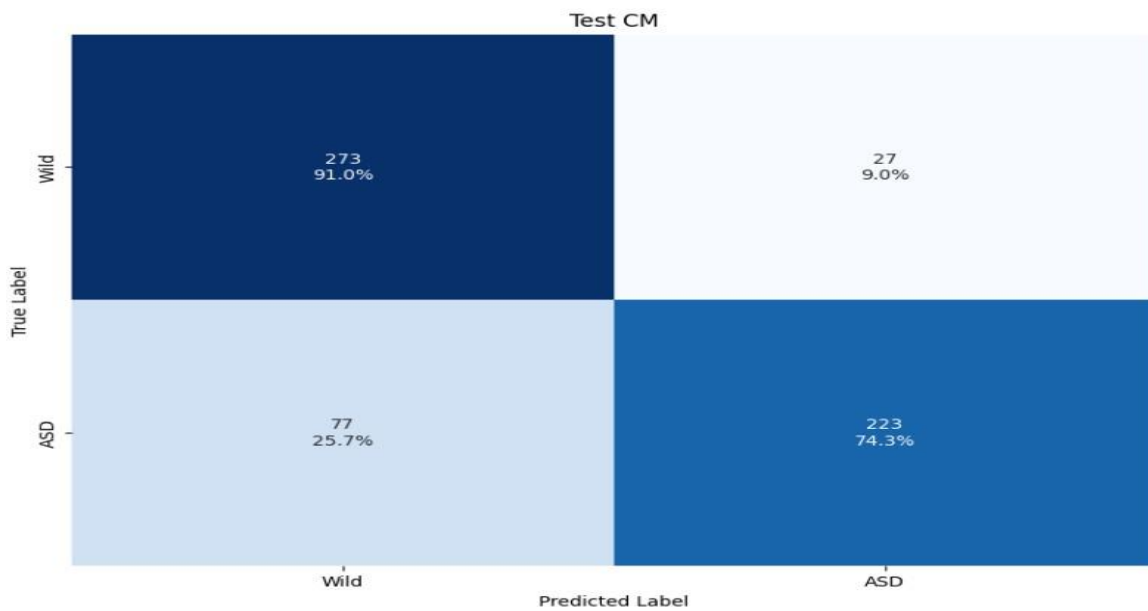


Figure 3 - PANNs learning curve for mother gene

The learning curve for classifying mother mice genes associated with ASD using the PANNs network, which incorporates mixup augmentation during training, demonstrates strong generalization performance. Validation accuracy surpasses training accuracy after around 200 iterations and remains consistently higher, reflecting the effectiveness of mixup augmentation in preventing overfitting. Both accuracies stabilize near 0.95, with the optimal performance observed around 800 iterations. The chosen number of estimators is 900, where the curve shows no further improvement, indicating that increasing iterations beyond this point does not enhance the model's accuracy, confirming the optimal balance between model complexity and generalization.

Table 4 - PANNs test results of mother gene



The model is more accurate in identifying Wild mice than detecting mASD, with a higher TN (91%) but a lower TP (74.3%). The significant FN (25.7%) indicates a frequent failure to detect ASD, while the FP (9%) suggests a relatively low tendency to over-diagnose ASD. This highlights challenges in balancing sensitivity and specificity, with a trade-off between correctly identifying ASD and minimizing false alarms.

4.5.2 Infant Gene Results

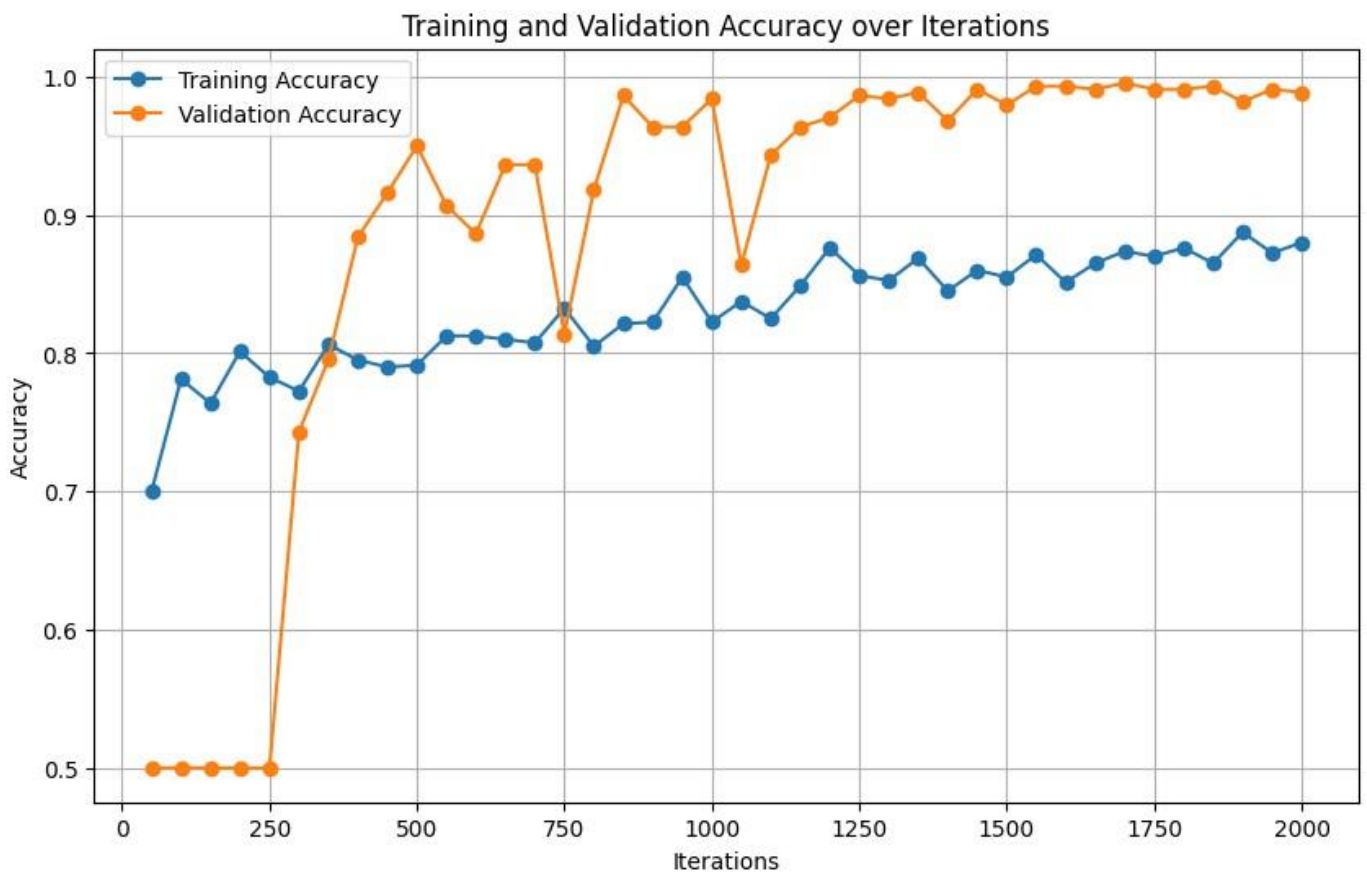
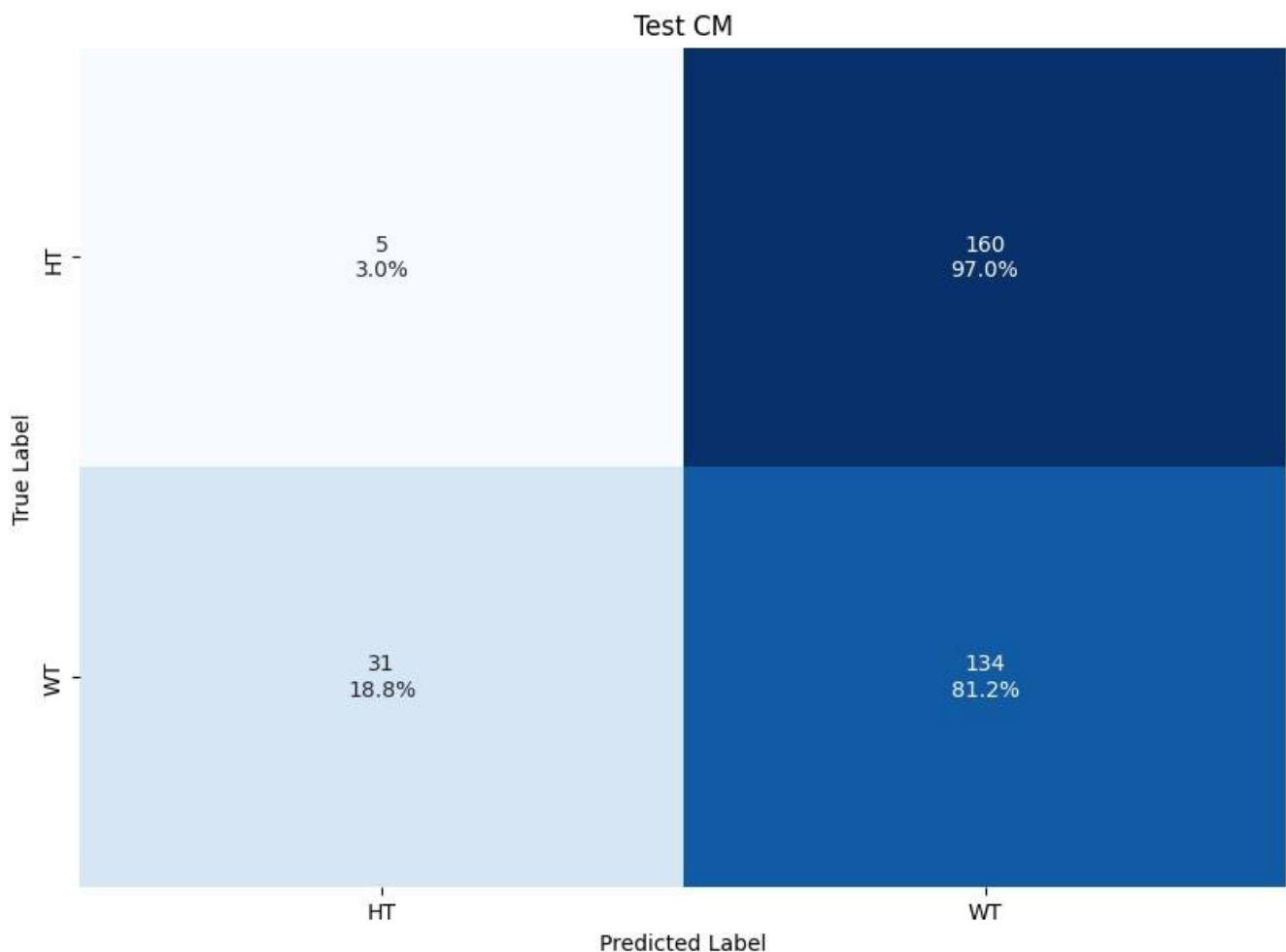


Figure 4 - PANNs learning curve for infant gene

The learning curve for classifying infant mice genes associated with ASD using the PANNs network, with mixup augmentation during training, reveals some concerning trends. While validation accuracy does surpass training accuracy around 250 iterations, suggesting good initial generalization, the overall pattern is less stable than expected. The validation accuracy fluctuates significantly before stabilizing near 0.95 around 1000 iterations, indicating potential issues with model consistency or the adequacy of the augmentation strategy. Moreover, despite reaching high validation accuracy, the training accuracy remains comparatively low and fails to show significant improvement after 500 iterations. This persistent gap suggests that the model might be underfitting the training data, potentially due to overly aggressive regularization or insufficient model capacity. The chosen number of estimators is 2000, but the lack of improvement beyond 1000 iterations raises questions about the effectiveness of extending training further, as it does not seem to provide any additional benefits to accuracy. Overall, while the model achieves reasonable performance, the instability and underfitting warrant further investigation and potential model adjustments.

Table 5 - PANNs test results of infant gene



The model demonstrates significant limitations in identifying ASD based on the infant gene. With a very low TP (3%) and a high FN (97%), the model struggles to correctly detect ASD in affected infants. Despite having a moderately high TN (81.2%), the model's FP (18.8%) suggests a tendency to over-diagnose ASD, indicating challenges in balancing sensitivity and specificity. These findings highlight the need for substantial improvements in the model to better capture the genetic markers of ASD in infants.

4.5.3 Three Gene Groups Results

In an attempt to refine the model's performance, the data was split into three groups based on gene type of the infant and the mother. This approach aims to achieve better classification results by providing the model with more specific genetic information. the first gene refer to the infant gen the second refer to the mother gene.

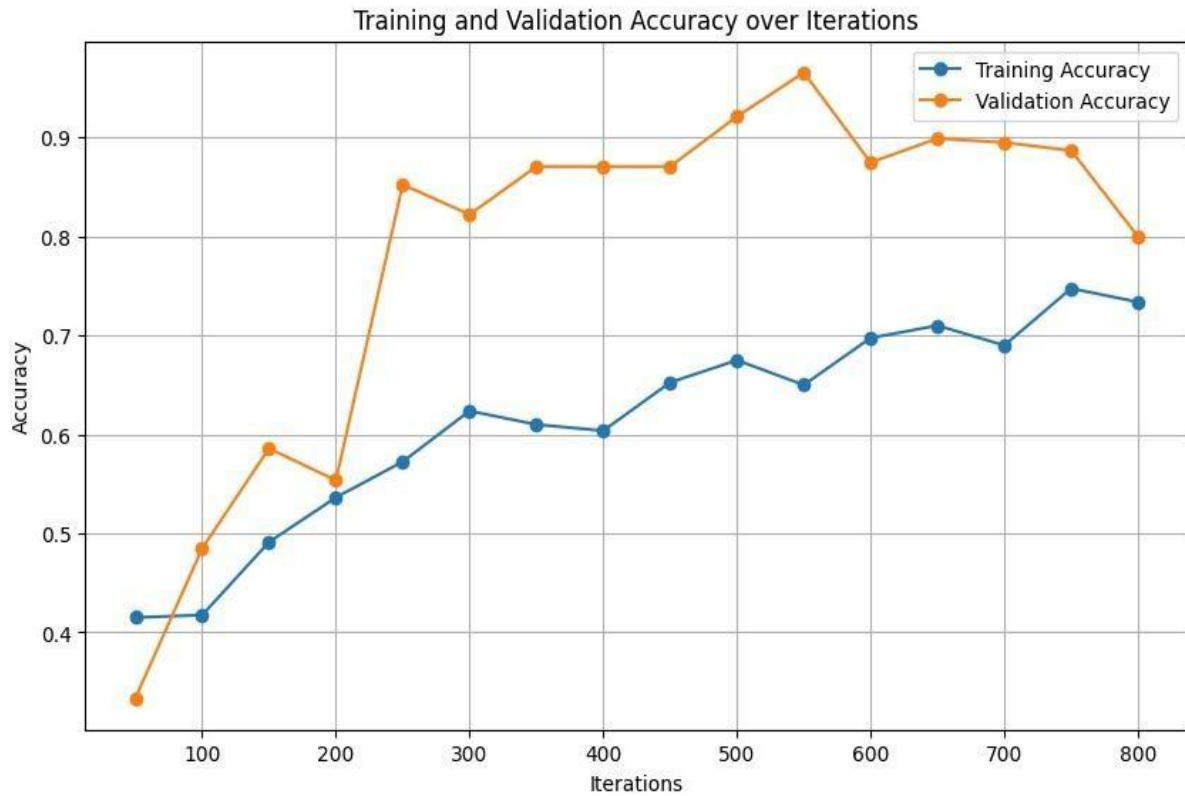


Figure 5 - PANNs learning curve for 3 gene group

The analysis of the learning curve for classifying the three classes of HT-WT, WT-WT, and WT-HT using the PANNs network with mixup augmentation reveals that 500 iterations is the most effective training point. Early in the process, validation accuracy surpasses training accuracy, but it begins to fluctuate and peaks around 500 iterations before declining, indicating that continuing training beyond this point may lead to overfitting. In contrast, training accuracy increases slowly and remains lower, suggesting potential underfitting

Table 6 - PANNs test results of three gene groups

		Test CM		
True Label	HT_HT	46 20.9%	76 34.5%	98 44.5%
	WT_HT	74 33.6%	145 65.9%	1 0.5%
	WT_WT	2 0.9%	4 1.8%	214 97.3%
		HT_HT	WT_HT	WT_WT
		Predicted Label		

The model is highly accurate in identifying WT_WT cases (TN = 97.3%) but struggles with HT_HT cases (TP = 20.9%), reflecting a significant difficulty in detecting ASD when both mother and infant are affected. The FN rate of 79.1% for HT_HT shows a frequent failure to detect ASD in these cases, while the FP rate of 34.5% between HT_HT and WT_HT suggests a tendency to misclassify cases where both have ASD as cases where only the mother has ASD. These results underscore challenges in the model's sensitivity and specificity across different genetic scenarios.

4.5.4 Discussion

4.5.1.1 Mother Gene

The PANNs model demonstrated impressive performance in detecting the mother gene associated with ASD, achieving an accuracy of 82%. This high level of accuracy reflects the model's effectiveness in identifying relevant genetic markers. Notably, the validation accuracy surpassed the training accuracy, likely due to the mix-up augmentation technique. This technique appears to have enhanced the model's ability to generalize from the training data and avoid overfitting, thereby improving performance on unseen validation data.

4.5.1.2 Infant Gene

The PANNs model performance with the infant gene was notably poor, with a true positive rate of only 3% for detecting ASD (HT) infants. This extremely low sensitivity highlights significant difficulties in accurately identifying ASD in infants based on genetic data alone. While the model achieved an accuracy of 81.2% in identifying non-ASD (WT) cases, the high false positive rate of 18.8% suggests a tendency to over-diagnose ASD. This over-diagnosis complicates the reliability of the model and underscores the challenge of distinguishing between ASD and non-ASD cases based on the infant's gene data.

4.5.1.3 Three-Group Gene

The PANNs model with the data segmented into three different gene groups aimed to improve model performance by incorporating a more detailed genetic dataset. However, this approach revealed that the model still faced challenges in consistently identifying ASD, especially in mixed gene-type scenarios. The complexity introduced by different gene combinations appeared to hinder the model's ability to accurately detect ASD, indicating areas where further refinement is necessary.

An initial issue with mouse detection due to a tagging bug was corrected, yet discrepancies between validation and test results persisted. These discrepancies highlight the need for further refinement in the model. The instability observed in the model's predictions suggests that while the approaches show potential, there are significant issues related to model reliability and generalization that require addressing.

5 Summary and conclusions

This project investigated multiple methodologies for detecting ASD in mice, including both traditional machine learning techniques like Majority Voting with XGBoost and advanced deep learning approaches using PANNs. The results revealed a mixed performance across different methods:

- Majority Voting with XGBoost: Initially demonstrated high accuracy but faced significant performance drops during testing due to data imbalance, limited test data, and augmentation limitations.
- Mother Gene with PANNs: Achieved high accuracy (82%) and benefited from mix-up augmentation, reflecting effective detection of maternal genetic markers associated with ASD.
- Infant Gene with PANNs: Struggled with a low true positive rate (3%) and a high false positive rate (18.8%), indicating difficulties in detecting ASD based on infant genes alone.
- Three-Group Gene: Provided a broader perspective but revealed challenges in consistent ASD identification across different gene types.

Overall, the study underscores the potential and limitations of the current methodologies in ASD detection. While promising results were achieved with the mother gene, there is a clear need for improvement in detecting ASD, particularly in cases involving infant genes and complex gene scenarios.

6 Future Work

To address the challenges identified and improve the model's performance, future research should focus on:

- **Analyze Result Discrepancies:** Investigate the reasons behind discrepancies between validation and test results, particularly in specific mice, to enhance model stability.
- **Dataset Segmentation:** Refine dataset segmentation strategies to better differentiate between healthy and ASD gene types, improving training and testing accuracy.
- **Potential for Improvement:** Explore training on a larger dataset and adjust model parameters to optimize performance. Consider alternative augmentation and regularization techniques to improve generalization.
- **Dataset Expansion:** Expand the dataset by collaborating with researchers to include a more diverse set of genetic samples. This expansion would allow the use of more sophisticated neural networks and deeper learning methods.
- **Parameter Tuning:** Conduct extensive experimentation with different model parameters and architectures to identify the optimal configuration for accurate ASD detection.

By focusing on these areas, there is potential to develop a more robust and reliable system for ASD detection, advancing our ability to diagnose and understand this complex condition in preclinical research settings.

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